Cover Letter

November 1, 2026

Faculty Search Committee
Department of Electrical and Computer Engineering
Princeton University
Princeton, New Jersey

Re: Tenure-Track Assistant Professor of Electrical and Computer Engineering, Interfolio 190824

Dear Members of the Search Committee:

I am writing to apply for the Assistant Professor position in Electrical and Computer Engineering at Princeton University. I earned a PhD in Electrical and Computer Engineering from the University of Oklahoma, where my research focused on making AI-based wireless systems more reliable when data are scarce, operating environments change, and models encounter conditions beyond their training. I would be excited by the opportunity to build an interdisciplinary research and teaching program at Princeton around reliable learning, communications, and intelligent sensing.

My doctoral research investigated data-efficient propagation modeling, generative and transfer learning for wireless environments, and AI-driven network outage detection and compensation. These studies combined communication-system models, simulation, experimental evaluation, and machine learning. Related work on smartwatch inertial sensing has broadened my interest in the reliability of learning from real-world measurements, particularly when devices and users differ. My research has been reported in peer-reviewed communications and intelligent-sensing venues, and my industry work at AT&T Labs, Futurewei, and AT&T has sharpened my understanding of practical measurement and system constraints.

I recognize that Princeton's current search gives particular emphasis to devices and device integration. My established expertise is not in semiconductor fabrication or materials processing; it is in the algorithms, sensing data, communication links, and systems-level decisions that can enable intelligent devices to perform reliably. I see a promising complementary research direction in collaboration with faculty studying integrated sensing and hardware systems: investigating how device variability, acquisition errors, communication constraints, and model uncertainty interact in end-to-end intelligent sensing. I would bring rigorous evaluation methods, data-efficient learning, and experience with signal and network measurements to those collaborations, while drawing on colleagues' device-physics and integration expertise.

My proposed program would also continue research in robust wireless prediction and networked autonomy. I would study when transfer or synthetic data improve predictions across deployment conditions and how uncertainty-aware methods can help systems respond to changing network states. Together with the sensing direction, these questions could support new collaborations among computing and networking, data and information science, materials and devices, and cyber-physical systems. I would build the program from reproducible software and measurement foundations, pursue external support suited to the science, and train students to treat careful evaluation as integral to invention.

I have taught Computer Programming, Communication Systems, and Wireless Communication to university undergraduates and supported engineering laboratory and research mentoring in earlier appointments. I would be prepared to contribute to teaching in communications, programming, signals, wireless and networked systems, and applied ML, with course assignments shaped by departmental needs. I value clear theoretical foundations, well-designed experiments, and mentoring that gives students increasing ownership of research questions.

Princeton's depth across devices, computing, sensing, and communications offers an especially compelling setting for this kind of collaborative research. I would welcome the opportunity to discuss both the contributions I could make immediately and the partnerships through which I could expand the reach of my work.

Thank you for your consideration.

Sincerely,

Waseem Raza, PhD

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